

Bywaf Usage Guide

Overview

Bywaf is a highly-auditable Python 3 commandlet framework for authorized web application and network testing workflows. It presents a Metasploit-like interactive shell, loads commandlets from plugins, and connects commandlets through a SQLite-backed event bus.

The core idea is simple: one commandlet discovers something and publishes it as an event; another commandlet consumes that event and publishes the next result. For example, **hostscanner** can publish live hosts, **portscanner** can consume those hosts and publish open ports, and HTTP commandlets can consume open ports and probe web services.

Use Bywaf only on systems and networks where you have explicit authorization.

Common task examples are collected in docs/FAQ.md. The documentation roadmap starts at docs/README.md. Evolving framework design notes are tracked in docs/DESIGN.md. Maintainer signing-key policy is recorded in docs/KEY_MANAGEMENT.md. Base capabilities, triggers, and audit/event topics are listed in docs/Framework_Surface.md. Core architectural references:

- docs/TERMINOLOGY.md defines jobs, pipelines, runs, events, topics, commandlets, plugins, capabilities, local IDs, and serials.
- docs/RUNTIME_MODEL.md explains runtime entities, lifecycle, foreground/background execution, control signals, and variable snapshots.
- docs/EVENT_MODEL.md explains event rows, topics, replay, framework requests, artifacts, notes, and provenance.
- docs/CAPABILITY_MODEL.md explains capability auditing, policy direction, and plugin integration types.
- docs/SYSTEM_BLOCK_DIAGRAM.pdf shows live runtime flow and durable data flow through the system.
- docs/SYSTEM_DATAFLOW_DIAGRAM.pdf focuses on command input, event, artifact, audit, request, and report data movement.

Installation

During development, run Bywaf from the repository root:

```
cd bywaf
python3 -m bywaf --help
python3 -m bywaf repl
```

For an editable local install:

```
python3 -m pip install -e .
```

```
bywaf --help
bywaf repl
```

For a local pip package build:

```
scripts/build_pip_package.sh
python3 -m pip install dist/bywaf-0.11.0-py3-none-any.whl
bywaf --help
```

For a local Debian package build, install the Debian build dependencies and write artifacts under `dist/deb/`:

```
sudo apt install debhelper dh-python pybuild-plugin-pyproject python3-all python3-setuptools
scripts/build_deb_package.sh
sudo apt install dist/deb/bywaf_0.11.0-1_all.deb
bywaf --help
```

For a local RPM package build, install RPM build tooling and write release artifacts under `dist/rpm/`:

```
sudo apt install rpm python3-build python3-installer
scripts/build_rpm_package.sh
```

Packaging smoke scripts:

```
scripts/build_release_packages.sh
tests/scripts/smoke_pip_package.sh
tests/scripts/smoke_rpm_package.sh
tests/scripts/smoke_plugin_install_paths.sh
tests/scripts/smoke_installed_package.sh
```

Release tags named `v*` trigger the GitHub Actions release workflow. The workflow builds pip, Debian, and RPM artifacts with the same local scripts, smoke-tests the package outputs, uploads the artifacts to the workflow run, and attaches them to the GitHub Release for the tag.

The project metadata defines a console script named `bywaf`, so packaged installations can expose Bywaf as a normal command instead of requiring `python3 -m bywaf`.

Bywaf can also be embedded as a Python library. A local GUI or web service should use the public session facade instead of scraping REPL output:

```
from pathlib import Path
from bywaf import BywafSession

session = BywafSession.open(Path(".bywaf/bywaf.sqlite3"))
session.run("hostscanner 127.0.0.1")
hosts = session.events(topic="host.found")
jobs = session.jobs()
```

The host and port scanner commandlets use `nmap` through a Python adapter. A local `nmap` binary is required for real scans. The adapter prefers `nmaplib`, then `python-nmap`, then `nmapthon`, then `libnmap`.

Dependency Summary

<code>nmap</code>	required for <code>hostscanner</code> and <code>portscanner</code>
<code>nikto</code>	required for the <code>nikto</code> wrapper commandlet
<code>eyewitness</code>	required for the <code>eyewitness</code> screenshot wrapper
<code>kismet</code>	required for the <code>wifi_scan</code> wireless wrapper
<code>prompt_toolkit</code>	required for rich interactive REPL completion
<code>libnmap/python-nmap/etc.</code>	Python <code>nmap</code> adapter; <code>Bywaf</code> tries supported adapters
<code>sqlcipher3-binary</code>	optional Python <code>SQLCipher</code> driver for encrypted DBs
<code>sqlcipher</code>	optional system <code>SQLCipher</code> tooling/library
<code>cryptography</code>	optional for signing-key management and bundle signing
<code>python-docx/openpyxl</code>	optional <code>DOCX/XLSX</code> table rendering backends
<code>dnspython</code>	optional for <code>dns_lookup</code>
<code>impacket</code>	optional for <code>smb_probe</code>
<code>ldap3</code>	optional for <code>ldap_probe</code>
<code>paramiko</code>	optional for <code>ssh_probe</code>
<code>pysnmp</code>	optional for <code>snmp_get</code>
<code>shodan</code>	optional for <code>shodan_lookup</code>
<code>yara-python</code>	optional for <code>yara_scan</code>

Install all dependencies with one of these command sets from the repository root.

Debian / Ubuntu:

```
sudo apt-get update
sudo apt-get install -y python3 python3-pip python3-venv python3-setuptools \
    python3-build python3-installer python3-prompt-toolkit nmap \
    sqlcipher libsqlcipher-dev debhelper dh-python pybuild-plugin-pyproject \
    python3-all rpm nikto kismet
python3 -m pip install -e '.[plugins,reporting,signing,sqlcipher]'
python3 -m pip install libnmap python-nmap nmapthon
```

Fedora / RHEL-family:

```
sudo dnf install -y python3 python3-pip python3-setuptools python3-build \
    python3-installer python3-prompt-toolkit nmap sqlcipher sqlcipher-devel \
    rpm-build nikto kismet
python3 -m pip install -e '.[plugins,reporting,signing,sqlcipher]'
python3 -m pip install libnmap python-nmap nmapthon
```

Use `yum` instead of `dnf` on older RHEL-family systems. Some external-wrapper tools, especially `eyewitness`, `nikto`, and `kismet`, may require a security-focused distribution, EPEL, or an upstream installation source.

Bywaf plugins are intended to wrap useful external tools and normalize their results into the central event database. That removes manual handoffs such as copying hosts to notes or intermediate files: one plugin can discover hosts, another can consume those host events, and later plugins can continue the workflow from the same stored data.

Execution-time plugin variables are scoped by commandlet. A plugin uses `context.vars.get("name")` for its own variables and cannot enumerate another plugin's variables through that API. Explicit global variables use `context.vars.get_global("name")`. When a commandlet run starts, Bywaf snapshots the effective commandlet and global variables into SQLite under that `command_run_id`; `event run=<id>` displays the captured variables so runs remain auditable and reproducible even when session variables change later. Runtime entities have two identities: local IDs for interactive typing (`job=12, run=1, pipeline=2`) and durable serials for audit/provenance. Local IDs are stable inside the current database and are never reused there, but they are not portable across replay/import into another database. Use `event serial=<serial>` when you want to inspect by the durable identifier. Explicit `load plugin=...` and `load script=...` operations also receive resource serials, so the load itself and the script commands it executed can be reviewed later.

Plugins that need interpreter-owned actions use request events instead of direct method calls. For example, a plugin can publish `shell.prompt.requested`; the foreground REPL validates the request and records either `shell.prompt.updated` or `framework.request.denied` for auditability. Plugins also declare intended capabilities on `CommandSpec`; Bywaf records audit-only `plugin.capability.used` and `plugin.capability.missing` events so operators can compare intended behavior with actual behavior. Plugin event-bus access should go through `context.events`, which audits `db.read:<topic>` and `db.write:<topic>` capability usage. Raw `context.db` access is retained for privileged/internal framework commandlets during the transition and audits `db.raw`.

Encrypted databases require SQLCipher support. On Debian or Ubuntu, install the SQLCipher library and use the optional Python extra:

```
sudo apt install sqlcipher libsqlcipher-dev
python3 -m pip install -e '.[sqlcipher]'
```

Starting Bywaf

Start the interactive shell:

```
bywaf
```

or, from a source checkout:

```
python3 -m bywaf
```

Create or open the default database with SQLCipher encryption:

```
bywaf --encrypt  
bywaf --database client.sqlite3 --encrypt
```

Run one command non-interactively:

```
bywaf run ls  
bywaf run cat README.md  
bywaf run 'hostscanner 127.0.0.1 | portscanner'
```

Simple `run` commands do not need quotes. Use quotes when the command contains shell metacharacters such as `|`, `&`, `>`, or spaces that must be preserved inside a single argument.

REPL Basics

The REPL prompt is:

```
bywaf>
```

Commandlets can be run directly:

```
bywaf> ls  
bywaf> cat README.md  
bywaf> hostscanner 127.0.0.1
```

Built-in commands manage the shell and the event database:

```
help  
help <command>  
plugins  
cmds  
vars  
history  
info  
job <list|show|cancel|end|kill>  
pipeline <list|show|cancel|end|kill>  
signal <job=id|run=id|serial=id> <action> [--soft|--hard] [key=value ...]  
cancel <job=id|pipeline=id|run=id>  
end [--soft|--hard] <job=id|pipeline=id|run=id>  
kill [--soft|--hard] <job=id|pipeline=id|run=id>  
jobs  
runs  
events [tail|--tail] [last=N]  
topics  
db <status|path|checkpoint|vacuum|new|encrypt|decrypt|rekey>  
event <topic|job=id|run=id|pipeline=id|serial=id>
```

```
load <resource>
save <resource>
exit
```

`help <command>` shows the same help as `<command> --help` for commandlets. Ctrl-C in the interactive shell asks whether to quit; answering yes exits through the normal shutdown path, including the SQLite checkpoint.

Commandlets

A commandlet is an executable unit exposed by a plugin. Each commandlet declares its name, description, arguments, consumed topics, and emitted topics.

Examples:

```
bywaf> help hostscanner
bywaf> hostscanner --help
bywaf> portscanner --help
```

Many scanning commandlets support `-s` or `--silent` to suppress console alerts. Commandlets request alerts through the framework using the database; the framework validates the request, stores a structured `console.alert` event, and prints it unless silent mode is active:

```
hostscanner <hostscanner-...>: discovered host 127.0.0.1
portscanner <portscanner-...>: discovered port 127.0.0.1:80/tcp
```

Commandlets can also declare tab-completion behavior for their arguments and options. For example, `ls [path]`, `cat <path>`, and `less <path>` get filename completion because those commandlets declare path/file completion in their plugin specs. Other completion specs include `topic`, `run`, `pipeline`, `job`, and `plugin`, so plugin authors can make hand-typed commands much easier to complete correctly. Runtime entity completions include prompt-toolkit metadata when available, such as serial, status/source, event counts, and the current number of attached artifacts.

Interactive shells use `prompt_toolkit` when a real terminal is available. **Ctrl-Space** enters completion-selection mode by opening the menu and selecting the first candidate. Then arrow keys move through candidates, **Enter** selects the highlighted completion, and **Esc** returns to the command line. A bottom toolbar shows that hint while a completion menu is open. The selection-mode key is configurable with `vars completion.select-key=<key>` using prompt-toolkit key names, because some desktop environments or terminal stacks reserve **Ctrl-Space**. `vars completion.wasd-selection=true` enables optional WASD-style menu navigation (**w/a** move backward, **s/d** move forward, following prompt-toolkit's flat completion order), but it is off by default so ordinary typing is not intercepted. Minimal non-interactive environments fall back to readline-style completion.

Plugin authors should use `context.output()`, `context.table()`, `context.alert()`, `context.progress()`, `context.page_file()`, and `context.process` instead of direct `print()` calls, direct terminal control, or direct subprocess calls. These helpers keep terminal output, progress, and external tool execution auditable and make the same commandlets usable from a future GUI or web frontend.

Progress is separate from findings. A finding is durable evidence such as `host.found` or `port.open`; progress is operational state such as “42% through the TCP scan.” Commandlets report progress through structured events:

```
context.progress_started(phase="tcp_scan", total=1000, unit="ports")
context.progress(phase="tcp_scan", current=420, total=1000, unit="ports")
context.progress_completed(phase="tcp_scan", current=1000, total=1000, unit="ports")
```

Bywaf enforces progress throttling in the framework. Configure it with global session variables:

```
bywaf> vars global.progress.min-interval-ms=250
bywaf> vars global.progress.min-percent-delta=1
```

Audit logs are stored as SQLite events. Use `audit show ...` to inspect them and `audit export file=audit.jsonl`, `audit export file=audit.pdf`, or `audit export file=audit.sqlite3` to hand off a copy.

```
bywaf> audit show topic=console.alert since=20260517 until=20260518
bywaf> audit export file=audit.pdf since=run=<command-run-id>
bywaf> audit export --encrypt file=audit.sqlite3
bywaf> audit export --encrypt file=audit.pdf
```

Unqualified `since=` and `until=` audit bounds default to `time:.` Encrypted SQLite audit exports use SQLCipher. Encrypted PDF export uses `pikepdf` when available, otherwise the external `qpdf` command.

Signing Keys

Bywaf keeps user signing and verification keys under `~/.bywaf/keys`, outside project databases. Private keys are encrypted with a passphrase, and signing operations ask for that passphrase instead of storing it in the project.

```
bywaf> key list
bywaf> key generate name=firm-evidence
bywaf> key show name=firm-evidence
bywaf> key import public file=reviewer.pub name=reviewer
bywaf> key export public name=firm-evidence file=firm-evidence.pub
bywaf> key test name=firm-evidence
```

`key list` shows computed signing state from the key files themselves: `available`, `locked`, `verify-only`, or `invalid`. The framework does not trust a user-edited

`can_sign` flag in metadata. Future signed bundle and audit-export commands will complete `key=` values from this keyring.

Evidence Bundles

Bundles collect audit records, evidence artifacts, and report artifacts into a durable manifest. The bundle definition is stored in the event DB as `bundle.created`, `bundle.item.added`, `bundle.sealed`, and `bundle.exported` events. Sealing hashes the current bundle contents; `--sign key=...` also signs that hashable manifest with a key from `~/.bywaf/keys`.

```
bywaf> bundle create name=client-a
bywaf> bundle add name=client-a audit since=20260501 until=20260519
bywaf> bundle add name=client-a evidence commandlet=nikto,webfin
bywaf> bundle seal name=client-a --sign key=firm-evidence
bywaf> bundle verify name=client-a
bywaf> bundle export name=client-a file=client-a.bundle.json
```

For artifact-backed bundle items, `commandlet=` accepts comma-separated commandlet names. The first implementation exports JSON bundles with artifact bodies encoded as Base64 and verifies signatures against the keyring. Sealed bundles reject additional `bundle add` operations; create a new bundle when more material needs to be added after sealing.

Plugins

A plugin provider groups related commandlets. The `plugins` command lists loaded providers:

```
bywaf> plugins
discovery
http
network
os
runtime
storage
```

The `cmds` command lists commandlets grouped by provider:

```
bywaf> cmds
discovery
  hostscanner
http
  eyewitness
  http_headers
  http_probe
  nikto
```



```

    webfin
network
    portscanner
os
    cat
    less
    ls
runtime
    job
storage
    db

```

Bundled plugins are listed in `bywaf/plugins/plugins.toml`. Adding a plugin file is not enough to load it by default; add its dotted path to that config and add or update its sidecar manifest, such as `bywaf/plugins/http/nikto.plugin.toml`.

External filesystem plugins are arbitrary local Python code. Bywaf refuses to load them unless plugin catalog trust is verified or the operator explicitly allows an unsigned development plugin. Use `--force` only when you want to bypass every plugin trust check for reviewed local code:

```
bywaf> load --force plugin=myplugin
```

Load a plugin from an explicit filesystem path:

```
bywaf> load --force plugin=./plugins/myplugin
bywaf> load --force plugin=~/.bywaf-plugins/myplugin
```

Startup plugin roots use the same policy. If you start Bywaf with `--plugin-root` and `--plugin-config`, use `--allow-unsigned-plugins` for unsigned development plugins:

```
bywaf --plugin-root ~/.bywaf/plugins --plugin-config ~/.bywaf/plugins/plugins.toml --allow-unsigned-plugins
```

Filesystem catalog builds use the same entry layout as runtime loading. A single-segment config entry such as `default_plugins = ["myplugin"]` points to `~/.bywaf/plugins/myplugin/plugin.py` and `~/.bywaf/plugins/myplugin/bywaf.plugin.toml`.

For reviewed external plugin trees, build and sign a catalog, then provide the catalog and trusted public key at startup:

```
bywaf --plugin-root ~/.bywaf/plugins \
  --plugin-config ~/.bywaf/plugins/plugins.toml \
  --plugin-catalog ~/.bywaf/plugins/plugin-catalog.signed.json \
  --plugin-catalog-key ~/.bywaf/plugins/plugin-catalog.pub.pem
```

Runtime catalog trust decisions are audited with `plugin.catalog.verified`, `plugin.catalog.rejected`, `plugin.catalog.entry.verified`, and `plugin.catalog.entry.rejected`.

`--allow-missing-plugin-keys` and `--allow-mismatched-plugin-keys` are narrower developer bypasses for future signed external plugin catalogs when the trusted verification key is absent or does not match the plugin

signature. `--plugin-manifest-key` supplies the trusted public key for signed `bywaf.plugin.toml` files. `--allow-unsigned-plugin-manifests` is the narrow development bypass for unsigned manifests. The legacy `--force-plugins` startup flag is a hidden compatibility alias for `--allow-untrusted-plugins`, which states the full tradeoff directly: load the plugin even though Bywaf cannot verify its signature, signing key, or key match. Official release public keys are reserved for `bywaf/keys/`; private signing keys are maintainer release material and must stay outside the repository and built packages. Official manifest-signing keys rotate annually with a staggered 60-day transition: publish the next public key, temporarily trust both keys, switch signing to the next key, re-sign and release official plugin manifests, then retire the old key. Revocation is reserved for suspected compromise or emergency distrust. Use `--plugin-manifest-key` to trust a local or third-party public key.

Pipelines

Pipelines connect commandlets with `|`. Prefix the expression with `name:` to name the pipeline without consuming commandlet arguments:

```
bywaf> hostscanner 127.0.0.1 | portscanner
bywaf> client subnet scan: hostscanner 127.0.0.1 | portscanner
```

The runner executes each stage in order. Events emitted by one stage are passed to the next stage as input. Events are also stored in SQLite with a pipeline ID and command run ID.

This model allows downstream commandlets to consume only the output relevant to the current pipeline, rather than every historical event in the database.

Attach a new background commandlet to an existing pipeline:

```
bywaf> pipeline attach <pipeline-id> portscanner run=<producer-run-id> since=beginning
bywaf> pipeline attach <pipeline-id> http_probe since=now
```

The attach selectors are orthogonal:

- `<pipeline-id>` chooses the pipeline the new command run joins.
- `run=<producer-run-id>` optionally narrows input to one upstream producer run.
- `since=beginning` replays matching historical events, then listens for new events.
- `since=now` ignores historical events and starts from the current event high-water mark.

If `run=` is omitted, the attached commandlet reads matching events from the whole pipeline.

Runtime Names

Name the current command run with a stage-local **name=** selector:

```
bywaf> hostscanner 127.0.0.1 name=localhost sweep
```

Name or inspect runtime entities after they exist:

```
bywaf> name run=<command-run-id> localhost sweep
bywaf> name pipeline=<pipeline-id> client subnet scan
bywaf> name job=<job-id> background listener
bywaf> name run=<command-run-id>
```

The explicit keyed form is **text=**, for example **name run=<id> text=localhost sweep**.

Assigned names appear in **runs**, **pipelines**, and **jobs** listings.

Framework Notes

Any commandlet stage can include a framework-level **note=** selector. The runner strips it before the commandlet receives arguments and records an audited **note.attached** event with the job, pipeline, and command-run IDs.

```
bywaf> hostscanner 10.0.0.0/24 note=client-approved internal subnet
```

If **note=** is the last selector in a stage, it consumes the rest of that stage without requiring quotes:

```
bywaf> hostscanner targets note=scope approved | portscanner note=top ports
```

Review attached notes with the **note** commandlet. Output and file exports use timestamp-first lines:

```
bywaf> note run=<command-run-id>
bywaf> note pipeline=<pipeline-id>
bywaf> note job=<job-id> file=notes.txt
bywaf> note add run=<command-run-id> text=follow-up note
```

Notes are append-only. Adding another note creates another timestamped **note.attached** event instead of replacing earlier notes.

Artifacts

Artifacts are evidence files attached to a run, pipeline, or job. Artifact bodies are stored in a separate artifact database next to the main database. If the main database is encrypted, the artifact database is encrypted with the same session passphrase. If the main database is plaintext, the artifact database is plaintext too. The main database stores timestamped provenance events such as **artifact.attached** and **artifact.exported**; it does not store artifact bodies.

Bywaf derives the artifact DB path from the active main DB path so the two files remain an integrity pair; arbitrary artifact DB switching is intentionally not exposed by default.

Start Bywaf with an encrypted database when you want SQLCipher-protected artifact bodies:

```
bywaf --encrypt
```

Attach one or more files:

```
bywaf> artifact attach run=<command-run-id> file=snapshot.html name='Landing page'
bywaf> artifact attach serial=<run-or-pipeline-or-job-serial> file=snapshot.html
bywaf> artifact attach run=<command-run-id> file=snapshot.html file=headers.txt
bywaf> artifact attach pipeline=<pipeline-id> file=report.json note=initial report
```

List, search, save, and verify artifacts:

```
bywaf> artifact list run=<command-run-id>
bywaf> search run=<command-run-id> name=landing
bywaf> search run=<command-run-id> filename=snapshot.html
bywaf> search run=<command-run-id> content=csrf
bywaf> artifact search run=<command-run-id> --regexp filename='.*\\.png'
bywaf> artifact replace artifact=1 file=snapshot-v2.html
bywaf> artifact remove artifact=1
bywaf> artifact save artifact=1 file=snapshot.html
bywaf> artifact save serial=<artifact-serial> file=snapshot.html
bywaf> artifact save run=<command-run-id> dir=artifacts/
bywaf> artifact verify pipeline=<pipeline-id>
```

Use **name=** for human-readable artifact labels; if omitted, the source filename is used. The **search** commandlet, and its **artifact search** alias, search artifact metadata quickly. **name=**, **filename=**, **note=**, and **content=** narrow the search to artifact names, source filenames, notes, or decoded text contents. Add **--regexp** to treat those field values as Python regular expressions. Any commandlet whose main action is text search should follow the same **--regexp** convention. **since=** and **until=** restrict matches by artifact creation time. Use **file=** when saving exactly one artifact. Use **dir=** when saving a set. If **file=** matches multiple artifacts, Bywaf reports that clearly and asks you to use **dir=** instead. For **artifact attach**, **serial=** may refer to a run, pipeline, or job serial. Artifact serials identify existing artifact rows for listing, searching, saving, and verifying; artifacts are not attached to other artifacts.

At-File Arguments

Any commandlet argument can use framework-level at-file expansion. Expansion happens before the commandlet receives its argument list and is audited as a framework argument expansion. When the active database is encrypted and

artifact storage is available, the expanded input file is also attached as an encrypted provenance artifact.

```
bywaf> hostscanner @lines:targets.txt
bywaf> hostscanner 192.168.1.1-255 except=@lines:do-not-scan.txt
bywaf> http_probe @target.txt
bywaf> echo @@literal-at-sign
```

Supported forms:

- **@file:** read the file as one text argument
- **@raw:file:** read the file as one text argument
- **@lines:file:** expand each non-empty line into a separate argument
- **@@value:** pass @value literally

Scanner commandlets support **except=** for exclusion lists. Values may be comma separated or file-backed through **@lines:.**

Tab completion preserves these prefixes while completing filesystem paths.

Variable Expansion

Bywaf expands **\$variables** before commandlets receive their argument list. Unquoted and double-quoted variables expand; single-quoted variables are passed literally.

```
bywaf> use hostscanner
bywaf> vars targets=192.168.1.1 192.168.1.2
bywaf> hostscanner $targets
bywaf> hostscanner "$targets"
bywaf> hostscanner '$targets'
```

Resolution checks the exact variable name first, then the active commandlet scope, then **global..** For example, **\$targets** in **hostscanner** checks **targets**, **hostscanner.targets**, and **global.targets**. Variable expansion is audited as **framework.variable.expanded**.

Plans And Policy

Commandlets can expose a framework-owned plan hook. Use **--test** to show the intended action and exit without running, or **--yes** to approve a required plan non-interactively:

```
bywaf> hostscanner 192.168.1.0/24 --test
bywaf> hostscanner 192.168.1.0/24 --yes
```

Plans are audited as **plan.requested**, evaluated as **policy.evaluated**, and approved or denied as **plan.approved** / **plan.denied** with **approved_by=<os**

user>. Suggested repairs, such as pruning out-of-scope targets for one run, are audited as `plan.repair.applied` or `plan.repair.denied`.

Initial network policy variables:

```
bywaf> vars global.policy.network.allow=192.168.1.0/24
bywaf> vars global.policy.network.deny=169.254.169.254/32,192.168.1.50
bywaf> vars global.plan.required=true
```

The policy layer applies to the run being launched. Repairs do not mutate source files, saved variables, or command history.

Command Continuation And Sequences

Use a trailing backslash to continue a command across physical lines in the interactive shell or in scripts:

```
bywaf> hostscanner \
... 192.168.0.0/24
```

Separate multiple commands with semicolons when they should run sequentially:

```
bywaf> vars target=127.0.0.1; vars; topics
```

Semicolons inside quotes are preserved as part of the argument text.

Background Execution

Append `&` to background a commandlet or pipeline:

```
bywaf> hostscanner 192.168.0.1-255 &
```

Normal commandlet execution is job-audited through the database. Foreground commandlets run in-process but still record `job.requested`, `job.claimed`, `job.started`, and `job.finished` or `job.failed`. Background jobs use the same job lifecycle, but a worker process claims and runs the queued job. Foreground management commands such as `db ...` and `job ...` run directly.

Stage-level backgrounding works inside pipelines:

```
bywaf> hostscanner 192.168.0.1-255 & | portscanner &
```

In that example, `portscanner` listens for `host.found` rows created by the immediately upstream `hostscanner` run in the same pipeline. It does not consume unrelated `host.found` rows from older scans.

A pipeline groups one or more runs in the same command expression or attached workflow. A run is one commandlet invocation inside that pipeline, such as the specific `hostscanner` stage or `portscanner` stage. A job is the supervised foreground/background execution lifecycle that runs one or more of those commandlet invocations. Operationally, jobs are chained together into

pipelines by the runs they supervise; one job may contribute the whole chain, or multiple jobs may contribute runs when commandlets are attached later. See `docs/TERMINOLOGY.md` for the canonical definitions of jobs, pipelines, runs, local IDs, serials, events, and topics.

Show the currently active runtime entities:

```
bywaf> info
```

List active jobs, or all jobs with an explicit active marker:

```
bywaf> job list
bywaf> job list --all
bywaf> job list --page
bywaf> jobs --all
```

Show one job:

```
bywaf> job show <id>
```

Soft-cancel a job so commandlets that check cancellation can exit cleanly:

```
bywaf> job cancel <id>
```

End a job. By default this is cooperative, like `cancel`; add `--hard` to force-stop the process:

```
bywaf> job end <id>
bywaf> job end --hard <id>
bywaf> job kill --hard <id>
```

Pipelines can be inspected and controlled the same way:

```
bywaf> pipeline list
bywaf> pipeline list --all
bywaf> pipeline list --page
bywaf> pipeline show <id>
bywaf> pipeline cancel <id>
bywaf> pipeline end <id>
bywaf> pipeline kill --hard <id>
```

`job list`, `runs`, and `pipeline list` show active runtime state by default. Use `--all` to include historical entries. These commands render table views with local ID, durable serial, lifecycle state, names, timestamps, and an `ARTIFACTS` column counting artifacts attached so far. Set `vars global.listing.active-format=long` to include the state timestamp in the state column; set it to `short` for compact lifecycle labels. Use `--page` on list actions such as `job list`, `pipeline list`, and `artifact list` to view long output through the framework pager.

`watchdog` is Bywaf's default service-style runtime monitor. Its plugin also provides a framework trigger:

```
ON plugin.capability.used capability=network.connect job_id=<active job>
DO watchdog --session-service
```

Interactive sessions evaluate provider-owned triggers and stop any session-scoped services during orderly shutdown. You can also run `watchdog` manually to tune thresholds or test the current DB:

```
bywaf> watchdog --once
bywaf> watchdog interval=10 timeout=300 stall-threshold=120 error-threshold=10 &
```

The `watchdog` emits `watchdog.timeout`, `watchdog.stalled`, and `watchdog.error_rate` events when active jobs exceed the configured limits. Trigger lifecycle events are also auditable: `framework.trigger.enabled`, `framework.trigger.fired`, and `framework.trigger.disabled`. Fired events include the source event ID that caused the trigger to fire. Use `triggers` to list loaded provider-owned trigger rules and their current cursors.

For live runtime control, `signal` is the canonical command for a concrete receiver: a job, a command run, or a `serial=` that resolves to one of those. A pipeline is a grouping scope, not executing code, so it does not receive plugin-domain signals directly. Use pipeline-aware commands such as `pause pipeline=...` or `end --hard pipeline=...` when you want the framework to fan out control over jobs associated with a pipeline. Framework-native signals such as `pause`, `resume`, `stop`, `end`, and `kill` apply the existing framework controls; plugin-domain signals such as `prune`, `mute`, `unmute`, and `verbosity` are delivered for commandlets to apply or ignore.

```
bywaf> signal run=<command-run-id> prune targets=192.168.1.0/24
bywaf> signal run=<command-run-id> mute
bywaf> signal serial=<run-or-job-serial> verbosity level=debug
bywaf> signal run=<command-run-id> verbosity level=debug
bywaf> signal run=<command-run-id> pause --hard
```

`cancel`, `end`, `kill`, `pause`, `resume`, and `stop` are convenience aliases over the same signal/control path. `end` and `kill` are synonyms; both default to cooperative `--soft`, and `--hard` force-terminates the affected process:

```
bywaf> cancel job=<id>
bywaf> cancel pipeline=<id>
bywaf> end job=<id>
bywaf> kill --hard pipeline=<id>
bywaf> pause job=<id>
bywaf> pause --hard job=<id>
bywaf> pause run=<command-run-id>
bywaf> resume --listonly pipeline=<id>
bywaf> resume --listonly run=<command-run-id>
bywaf> stop --hard job=<id>
```


Advanced Runtime Control

Most users should prefer the friendly commands above. **signal** is the explicit runtime-control form for debugging, plugin development, and rare control messages that do not yet have their own top-level verb. Its main value is auditability: the database records the exact control payload requested and the plugin or framework response.

For example, these friendly commands:

```
bywaf> pause run=7
bywaf> end --hard job=3
```

map to explicit control requests like:

```
bywaf> signal run=7 pause --soft
bywaf> signal job=3 end --hard
```

For plugin-specific messages, the explicit form is the normal route because the plugin owns the meaning of the action:

```
bywaf> signal run=7 prune targets=192.168.1.0/24
bywaf> signal run=7 verbosity level=quiet
```

The resulting audit trail shows the requested selector, action, strength (**--soft** or **--hard**), supplied arguments, and whether the receiver applied, ignored, or rejected the request.

jobs remains as a convenience alias for **job list**, and **pipelines** remains as a convenience alias for **pipeline list**.

Database and Event Model

Bywaf stores events in SQLite. The default database is:

```
.bywaf/bywaf.sqlite3
```

SQLite is used in WAL mode. Inserts are committed immediately; there is no separate document-style save step. On shutdown, Bywaf checkpoints the WAL using `PRAGMA wal_checkpoint(TRUNCATE)` so WAL contents are folded back into the main database file.

List known event topics:

```
bywaf> topics
```

Show the last 25 events, or choose an explicit tail size:

```
bywaf> events
bywaf> events tail
bywaf> events tail last=50
```

Show recent events for a topic:

```
bywaf> event host.found
bywaf> event port.open
```

List command runs:

```
bywaf> runs
```

Show events by command run or pipeline:

```
bywaf> event run=1
bywaf> event pipeline=1
bywaf> event serial=<durable-serial>
```

Save a database snapshot:

```
bywaf> save db=snapshot.sqlite3
```

Save an encrypted snapshot:

```
bywaf> save --encrypt db=snapshot.sqlite3
```

Inspect or maintain the active database:

```
bywaf> db status
bywaf> db path
bywaf> db checkpoint
bywaf> db vacuum
```

Create a fresh database and switch the active session to it:

```
bywaf> db new
bywaf> db new --file=client.sqlite3
bywaf> db new --encrypt --file=client.sqlite3
bywaf> db new --force --file=client.sqlite3
```

Without `--file`, `db new` creates a timestamped database under `.bywaf/db/`. With `--file`, it refuses to overwrite an existing file. Add `--force` to move the existing database and SQLite sidecar files to timestamped `.bak-*` names before creating the new DB. `--encrypt` forces SQLCipher encryption and prompts twice for a passphrase.

The session variable `db.encryption=sqlcipher` makes `db new` encrypted by default:

```
bywaf> vars db.encryption=sqlcipher
bywaf> db new
```

Convert the active database in place:

```
bywaf> db encrypt
bywaf> db rekey
bywaf> db decrypt
```

`db encrypt` converts the active plaintext database to SQLCipher and prompts twice for a new passphrase. `db rekey` changes the passphrase for an encrypted

database. `db decrypt` exports the active encrypted database back to plaintext SQLite after an explicit **YES** confirmation.

Switch to another database:

```
bywaf> load db=snapshot.sqlite3
```

If the database is encrypted, Bywaf prompts for its passphrase when loading it. Passphrases are kept in process memory only and are not written to config or history files.

Projects

A project is a named working directory with its own database, config, and history:

```
~/.bywaf/projects/<name>/bywaf.sqlite3
~/.bywaf/projects/<name>/config.toml
~/.bywaf/projects/<name>/history.bywaf
```

Start in an existing project, or create one before startup:

```
bywaf project=client-a
bywaf --new project=client-b
bywaf --new --encrypt project=client-c
```

Manage projects from the REPL:

```
bywaf> project list
bywaf> project info
bywaf> project new name=client-b
bywaf> project use name=client-b
```

Switching projects changes the active database, config, and history path. Bywaf refuses to switch while jobs are active because those jobs belong to the current database. To deliberately hard-stop active jobs and switch anyway:

```
bywaf> project use name=client-b --force
```

The forced stop is audited in the old project database before the switch.

Resource Files

Bywaf keeps default state in:

```
.bywaf/
```

Important files:

```
.bywaf/bywaf.sqlite3
.bywaf/config.toml
```

```
.bywaf/history.bywaf
.bywaf/plugins/
```

Resource resolution rules are consistent:

```
plugin=<name>    -> .bywaf/plugins/<name>
script=<name>    -> ./<name>
db=<name>        -> ./<name>
config=<name>    -> ./<name>
history=<name>   -> ./<name>
```

Explicit paths are used as filesystem paths for every resource type:

```
./name
../name
~/name
/absolute/name
```

Examples:

```
bywaf> load script=scan.bywaf
bywaf> load script=./scripts/scan.bywaf
bywaf> save config=session.toml
bywaf> load config=session.toml
bywaf> save history=session-history.bywaf
bywaf> load history=session-history.bywaf
```

Variables

List variables:

```
bywaf> vars
```

Set a variable:

```
bywaf> vars name=value
```

Secret-looking variable names such as `password`, `pw`, `token`, `secret`, `api-key`, `authorization`, and `cookie` are stored as secret references in the normal variable store. `vars`, command history, and audit-friendly displays show `<redacted>` plus an HMAC fingerprint instead of the plaintext:

```
bywaf> vars ssh_probe.password=client-password
ssh_probe.password=<redacted> fingerprint=hmac-sha256:...
```

Credential-aware commandlets resolve those references through the framework secret API at run time. This keeps the normal variable store redacted while still allowing commandlets such as `ssh_probe`, `ldap_probe`, `smb_probe`, and `shodan_lookup` to authenticate.

Secrets are persisted in the active database so they survive restart. If the database is encrypted, they are protected by that database encryption at rest. If the database is plaintext, Bywaf prints a warning before storing the secret there.

Show one variable:

```
bywaf> vars name
```

Common examples:

```
bywaf> vars http_probe.cookie-file=/tmp/cookies.txt
bywaf> vars history.timestamp-format=%Y-%m-%d %H:%M:%S %Z
bywaf> vars hostscanner.targets=192.168.1.1-255
bywaf> hostscanner
```

Use a commandlet context to make short variable assignments target that commandlet:

```
bywaf> use hostscanner
bywaf> vars targets=192.168.1.1-255
bywaf> use global
```

For commandlets that opt into variable defaults, explicit command-line arguments take precedence over commandlet variables, and commandlet variables take precedence over built-in defaults. For example, `hostscanner 127.0.0.1` ignores `hostscanner.targets`, while `hostscanner` falls back to it.

Save variables:

```
bywaf> save config=config.toml
```

Load variables:

```
bywaf> load config=config.toml
```

Config files are TOML tables containing session variables. Legacy JSON config files can still be loaded for compatibility.

History

Every non-empty REPL command is appended to:

```
.bywaf/history.bywaf
```

History lines are stored as commands followed by a timestamp comment:

```
hostscanner 127.0.0.1 # 2026-05-12 10:15:30 EDT
```

The `history` command shows only commands from the current REPL invocation, with timestamps displayed first for easier scanning:

```
bywaf> history
2026-05-17 10:00:00 EDT  plugins
```

Limit session history by timestamp with `since=` and `until=`. Unqualified values default to `time:` and use `yyyymmdd[HH[MM[SS]]]`:

```
bywaf> history since=20260517 until=20260518
bywaf> history since=time:202605171000 until=time:202605171059
```

The persistent history file can be viewed with the OS commandlets:

```
bywaf> cat .bywaf/history.bywaf
bywaf> less .bywaf/history.bywaf
```

The persistent history file stays script-friendly: timestamps are stored after commands as comments, so history lines can be copied into a script file.

Change the timestamp format:

```
bywaf> vars history.timestamp-format=%Y/%m/%d %H:%M:%S %Z
```

Scripts

Scripts are text files with one command expression per line:

```
# scan local host
hostscanner 127.0.0.1
portscanner --from-topic host.found
```

Blank lines and lines beginning with `#` are ignored. Inline comments are also allowed after whitespace:

```
plugins # list loaded plugin providers
```

Run a script:

```
bywaf> load script=scan.bywaf
```

Bundled Commandlets

os

`ls` lists local files:

```
bywaf> ls
bywaf> ls bywaf/plugins
```

`cat` prints a local text file:

```
bywaf> cat README.md
```

`less` opens the system `less` pager when running interactively:

```
bywaf> less README.md
```

Use `/` to search inside `less`, arrow keys or paging keys to scroll, and `q` to quit.

discovery

`hostscanner` discovers live hosts with `nmap`:

```
bywaf> hostscanner 127.0.0.1
bywaf> hostscanner 192.168.0.1-255
bywaf> hostscanner 192.168.1-3.1-255
```

It emits `host.found` events.

network

`portscanner` scans hosts for open ports:

```
bywaf> portscanner 127.0.0.1
bywaf> portscanner --ports 22,80,443 127.0.0.1
```

If `--ports` is omitted, `nmap` uses its normal default top-port behavior. It emits `port.open` events.

Use listen mode to consume newly inserted hosts:

```
bywaf> portscanner --listen
```

`ssh_probe` uses `Paramiko` for SSH service/auth probing and emits `ssh.service`:

```
bywaf> ssh_probe 127.0.0.1
bywaf> ssh_probe username=test password=test 127.0.0.1
```

`snmp_get` uses `pysnmp` to read one OID and emits `snmp.value`:

```
bywaf> snmp_get community=public oid=1.3.6.1.2.1.1.1.0 127.0.0.1
```

recon

`dns_lookup` uses `dnspython` and emits `dns.record` or `dns.error`:

```
bywaf> dns_lookup example.com
bywaf> dns_lookup record-type=MX example.com
```

`shodan_lookup` uses the Shodan Python library. Set `SHODAN_API_KEY`, use `api-key=...`, or set vars `shodan_lookup.api-key=...`

```
bywaf> shodan_lookup 8.8.8.8
bywaf> shodan_lookup mode=search apache country:US
```

identity

`ldap_probe` uses `ldap3` and emits `ldap.server`:

```
bywaf> ldap_probe dc.example.test
bywaf> ldap_probe username='EXAMPLE\\user' password=secret dc.example.test
```

`smb_probe` uses `Impacket` and emits `smb.server`:

```
bywaf> smb_probe 127.0.0.1
bywaf> smb_probe domain=EXAMPLE username=user password=secret dc.example.test
```

http

`http_headers` performs HTTP HEAD requests and emits `http.headers` events:

```
bywaf> http_headers example.com
bywaf> http_headers --ssl true example.com
```

`http_probe` probes HTTP endpoints and emits `http.endpoint` events:

```
bywaf> http_probe https://example.com/
```

`webfin` fingerprints HTTP endpoints and emits `web.fingerprint` events:

```
bywaf> http_probe https://example.com/ | webfin
bywaf> webfin https://example.com/
```

`nikto` wraps the Nikto web scanner through the framework process API, attaches raw JSON output as an artifact, and emits `nikto.finding`, `vulnerability.found`, and `vulnerability.potential` events:

```
bywaf> nikto https://example.com/
bywaf> http_probe https://example.com/ | nikto
bywaf> http_probe https://example.com/ | webfin | nikto
```

`source=webfin` is available when Nikto receives mixed upstream input and you want it to ignore plain `http.endpoint` events.

`eyewitness` wraps EyeWitness for web screenshots. It writes output under `.bywaf/eyewitness/<run-id>` by default, or under `--output-dir` when supplied. Screenshot files are also attached to the encrypted artifact store when the active database is encrypted, or the plaintext artifact store otherwise.

```
bywaf> eyewitness https://example.com/
bywaf> http_probe https://example.com/ | eyewitness
bywaf> eyewitness --output-dir=client-shots https://example.com/
```

For authorized session-aware testing, it can use cookies:

```
bywaf> vars http_probe.cookie-file=/path/to/cookies.txt
bywaf> http_probe https://example.com/
bywaf> http_probe --firefox-profile ~/.mozilla/firefox/<profile>
```

wireless

`wifi_scan` wraps a Kismet-style wireless scan. It writes logs under `.bywaf/wireless/<run-id>` by default, attaches produced files to the paired artifact store, and emits `wifi.network` plus `kismet.network` events from JSON output when present.


```
bywaf> wifi_scan interface=wlan0mon duration=60
bywaf> vars wifi_scan.interface=wlan0mon
bywaf> wifi_scan duration=120
```

analysis

`finding_dedupe` normalizes vulnerability/finding events and emits deduplicated finding lifecycle events: `finding.new`, `finding.duplicate`, `finding.updated`, and `finding.merge_candidate`. It preserves the original scanner output and points normalized findings back to their source events, so a reporter plugin can later render tables from the normalized event stream.

```
bywaf> nikto https://example.com/ | finding_dedupe
bywaf> finding_dedupe file=dedupe-summary.json
bywaf> finding_dedupe format=md file=findings.md
```

Exact matches use standardized identifiers such as CVE/CWE/GHSA/vendor IDs when available, then target identity and stable evidence fingerprints. Fuzzy text matching is only used as a low-confidence `finding.merge_candidate`.

`finding_report` renders normalized dedupe events or raw tool findings as a table through the framework table provider. The columns are Finding name, Description, Host(s) affected, CVE, Severity rating, and Recommendation. `export=` writes the table to a file, infers the format from the suffix, and attaches the report as an artifact.

```
bywaf> finding_report
bywaf> finding_report source=tools
bywaf> finding_report export=findings.md
bywaf> finding_report export=findings.xlsx
```

`yara_scan` uses yara-python and emits `yara.match`:

```
bywaf> yara_scan rule=webshells.yar shell.php
```

Common Workflows

Scan one host for live status:

```
bywaf> hostscanner 127.0.0.1
```

Scan one host for ports:

```
bywaf> portscanner 127.0.0.1
```

Discover hosts and scan their ports:

```
bywaf> hostscanner 192.168.0.1-255 | portscanner
```

Run discovery and port scanning in the background:

```
bywaf> hostscanner 192.168.0.1-255 & | portscanner &
```

Probe HTTP services after port scanning:

```
bywaf> hostscanner 127.0.0.1 | portscanner --ports 80,443 | http_probe
```

Fingerprint HTTP services after probing:

```
bywaf> hostscanner 127.0.0.1 | portscanner --ports 80,443 | http_probe --method GET | webfinger
```

Save the current database:

```
bywaf> save db=scan-results.sqlite3
bywaf> save --encrypt db=scan-results.sqlite3
```

Save variables and session history:

```
bywaf> save config=session.json
bywaf> save history=session.bywaf
```

Troubleshooting

If `python` is not available, use `python3`:

```
python3 -m bywaf
```

If scanning fails with an `nmap` error, verify that `nmap` is installed and that the selected Python `nmap` binding is available.

If socket creation is denied, the process may be running inside a sandbox or without the privileges needed by the selected scan type.

If a command is unknown, check loaded commandlets:

```
bywaf> cmds
```

If a plugin does not load, verify its directory structure and that it defines a `plugin()` factory returning a commandlet.

SQLite WAL files such as `.sqlite3-wal` and `.sqlite3-shm` are normal. Bywaf checkpoints the WAL on shutdown.

Developer Notes

The main package layout is:

<code>bywaf/app.py</code>	REPL and built-in commands
<code>bywaf/command_parser.py</code>	commandlet and pipeline parsing
<code>bywaf/runner.py</code>	pipelines, foreground/background execution
<code>bywaf/db.py</code>	SQLite event store
<code>bywaf/plugin.py</code>	commandlet protocol and specs
<code>bywaf/registry.py</code>	plugin discovery and loading
<code>bywaf/completion.py</code>	prompt_toolkit/readline completion

```
bywaf/plugins/      bundled plugin providers
tests/              unit tests
```

Run tests:

```
python3 -m unittest discover -s tests
```

Run completion regressions only:

```
python3 -m unittest discover -s tests -p 'test_completion_regression.py'
```

The completion regression suite loads every default commandlet and verifies that command names, binary `--flag` options, value-bearing `name=` arguments, choice values, filespec values, `$variable` references, runtime selectors, pipe-position command completion, and prompt-toolkit display labels stay consistent.

Run the PTY-level readline Tab smoke tests only:

```
python3 -m unittest discover -s tests -p 'test_interactive_completion_smoke.py'
```

Those smoke tests launch the real REPL under `pexpect`, type partial commands, send Tab, and verify the terminal text. They force `BYWAF_INPUT_READER=readline` for deterministic PTY behavior; prompt-toolkit display behavior is covered by the faster completion adapter tests.

Run user-facing script flows only:

```
python3 -m unittest tests.test_user_flows
python3 tests/scripts/run_user_flow.py tests/user_flows/basic_runtime.bywaf
```

User flows are ordinary `.bywaf` scripts with optional `# EXPECT:` and `# EXPECT-EVENT:` assertions. They exercise real REPL/script commands from the operator's point of view and double as executable examples.

Validate a filesystem plugin package outside the Bywaf interpreter:

```
python3 scripts/plugin_check.py path/to/plugin-dir
python3 scripts/plugin_check.py path/to/plugin-dir --manifest-key manifest-signing.pub.pem -
python3 scripts/plugin_check.py path/to/plugin-dir --json
python3 scripts/plugin_manifest_sign.py --manifest path/to/plugin-dir/bywaf.plugin.toml --pr
```

The maintainer keeps private manifest-signing keys outside the repository. Official public verification keys are packaged under `bywaf/keys/` when released. Official manifest-signing keys rotate annually with a 60-day staggered transition; official plugin manifests are re-signed and released with the new key, and old keys are retired after the transition window. Revocation is reserved for suspected compromise or emergency distrust. `--plugin-manifest-key` can point at another trusted public key.

Build and verify a maintainer-side signed plugin catalog:

```
python3 scripts/plugin_catalog.py build --output dist/plugin-catalog.json
python3 scripts/plugin_catalog.py sign \
    --catalog dist/plugin-catalog.json \
```

```

--private maintainer-plugin-signing.pem \
--signer "Bywaf maintainer" \
--output dist/plugin-catalog.signed.json
python3 scripts/plugin_catalog.py verify \
--catalog dist/plugin-catalog.signed.json \
--public maintainer-plugin-signing.pub.pem \
--check-tree

```

The catalog binds the reviewed bundled plugin list, plugin source hashes, sidecar manifest hashes, traits, commandlets, capabilities, and secret options. `--check-tree` proves the signed catalog still matches the files in the current checkout.

Add a commandlet by defining a class with a `CommandSpec` and a `run()` method, then expose it through a `plugin()` factory. Add bundled commandlets to `bywaf/plugins/plugins.toml` when they should load by default.

Commandlets can declare completion metadata with `ArgumentSpec`, `OptionSpec(..., completion=CompletionSpec(...))`, or an optional custom `complete(context, args, prefix)` method. See `docs/PLUGIN_AUTHOR_GUIDE.md` for a walkthrough and a small working example.

Reference

Useful built-ins:

```

help [command]
plugins
cmds
vars [name=value]
history
job <list|show|cancel|end|kill>
pipeline <list|show|cancel|end|kill|attach>
signal <job=id|run=id|serial=id> <action> [--soft|--hard] [key=value ...]
cancel <job=id|pipeline=id|run=id>
end [--soft|--hard] <job=id|pipeline=id|run=id>
kill [--soft|--hard] <job=id|pipeline=id|run=id>
name <run=id|pipeline=id|job=id> [name text]
note [add] <run=id|pipeline=id|job=id> [text=note|file=path]
artifact <attach|list|remove|replace|save|search|verify> [artifact=id|run=id|pipeline=id|job=id]
search [--regexp] <name=text|filename=text|note=text|content=text> [artifact=id|run=id|pipeline=id|job=id]
jobs
pipelines
runs
events [tail|--tail] [last=N]
topics
event <topic>

```

```
event job=<id>
event run=<id>
event pipeline=<id>
event serial=<id>
db <status|path|checkpoint|vacuum|new|encrypt|decrypt|rekey>
key <list|show|generate|import|export|remove|test>
bundle <create|add|list|show|seal|verify|export>
load [--force] plugin=<resource>
load script=<resource>
load db=<resource>
load config=<resource>
load history=<resource>
save db=<resource>
save --encrypt db=<resource>
save config=<resource>
save history=<resource>
prompt [pattern]
exit
```

Common event topics:

```
host.found
port.open
http.headers
http.endpoint
job.requested
job.claimed
job.started
job.finished
job.failed
```